

Lecture Summary: Exception Handling & Logging in Python

1. Exception Handling (The try-except-finally-else system) The goal of exception handling is to manage errors gracefully so that the entire system doesn't terminate unexpectedly.

- Try Block: This is where you place the code that might cause an error (e.g., dividing by zero or opening a missing file).
- Except Block: This code runs only if an error occurs. The lecture recommends using specific error classes (like `ZeroDivisionError` or `FileNotFoundError`) rather than a general catch-all.
- Exception Superclass: While specific errors are better, the `Exception` class can be used as a "safety net" to catch any error that wasn't specifically defined.
- Else Block: This runs only if the code in the `try` block was successful (no errors occurred).
- Finally Block: This block always runs, regardless of whether an error happened or not. It is typically used for "cleanup" tasks, like closing a database connection or a file.
- Nested Try: You can place a `try-except` block inside another `try` block for more complex error management.

2. Logging in Python The lecture advocates for the logging module over `print()` statements. Printing only shows information in the console while the app is running, but logging saves a permanent record to a file.

Logging Levels (from least to most severe):

1. DEBUG: Detailed information, typically of interest only when diagnosing problems.
2. INFO: Confirmation that things are working as expected.
3. WARNING: An indication that something unexpected happened, or a problem might occur in the near future.
4. ERROR: Due to a more serious problem, the software has not been able to perform some function.
5. CRITICAL: A serious error, indicating that the program itself may be unable to continue running.

3. Basic Configuration The `logging.basicConfig()` function is used to set up where the logs are saved (filenames) and which level of messages should be captured.

4. Practical Application The lecture demonstrates these concepts through a Calculator program. By using exception handling, the calculator can handle a user trying to divide by zero without crashing. By using logging, the developer can keep a history of all calculations and errors that occurred during the session.

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